

Information Efficiency in Life Sciences Equity Prices: Clinical Trial Disclosures vs. Earnings Announcements in San Diego

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Abstract

This paper asks whether equity prices absorb clinical trial results more slowly than quarterly earnings, on the reasoning that trial data is more technical, hence harder to interpret. Using 2,349 events from 60 San Diego life sciences firms between 2019 and 2025, with 8-K filings classified by an LLM pipeline validated against hand labels, we found that clinical announcements produce a significant same-day price reaction but no pre-announcement leakage, no post-announcement drift, and no detectable difference from earnings announcements. A placebo benchmark of 24,600 matched non-event dates shows clinical-stage firms decline 2.6% over arbitrary nine-day windows, so a conventional test against zero would have produced a spurious leakage finding. The null holds across five robustness specifications.

Introduction

Clinical trial results are one of the hardest disclosures for people to interpret in public markets. As an example, to comprehend a Phase 3 readout, a person has to know the definition of endpoints, the statistical plan, and the base rate of success in the indication. On the other hand, quarterly earnings announcements are highly standardized and they arrive with analyst forecasts. If the complexity of information affects price discovery rate, these two types of announcements should cause different reactions.

San Diego is one of the three major life sciences cities in the United States, alongside Boston and San Francisco. Out of 132 companies that are located in San Diego, 73 of which are life sciences firms, which is 55.3% of all the companies, a dense concentration in a single geography. By studying 60 life sciences firms in San Diego between 2019 to 2025, we found that clinical results provided a same-day price reaction, and there was no evidence on information leakage and price drift after announcement. Since the price move once and stop and react on the same day it comes out, we found no detectable difference between clinical trial disclosures and financial announcements.

Data

To construct the company universe, we used the firms' self-reported Standard Industrial Classification (SIC) code from SEC submissions API. The SIC code is a four-digit code that indicates the industry a company operates in. To start with, there were 7,997 companies with an active ticker in August 2026, and we retained the 132 companies that fall within San Diego County. Out of these companies, we classified 73 as life sciences companies by manual review.

SIC codes need to be manually reviewed because they are self-reported at the time of filing and not verified by the SEC. One example of wrong classification is Premier Air Charter Holdings, a private jet company that assigned itself under SIC code 3841, Surgical and Medical Instruments. The API also only returns the firms' current address, so we parsed the address of the companies from the cover page of the filings and dropped the companies with over half of the filings that are not located in San Diego. This leaves us 60 companies.

From these 60 companies, we extracted 4,788 8-K filings covering 2019-2025. An 8-K is a mandatory filing that firms need to submit once they undergo a material event, which makes it a legally updated record. However, since there is no category in 8-K for clinical results, the biotechnology catalysts are mostly under the filings labeled 7.01(Regulation FD Disclosure) and 8.01(Other Events). Both 7.01 and 8.01 filings contain lots of routine investment events, so we needed to read the texts in order to classify them. To make the classification more efficient, we built a LLM classification pipeline to label the filings as CLINICAL, REGULATORY, PRESENTATION and OTHER. 100 filings were hand labeled before LLM classification, and the output gave us a Cohen's kappa check of 0.817.

When an earnings release and a catalyst arrive in the same filing, the price reaction on that date cannot be attributed to either, so we excluded the 206 filings where this occurs. We then aligned each remaining filing to a trading day, assigning announcements filed after 16:00 Eastern to the next session.

Method

For each event, we estimated a market model based on 220 trading days: from $t-250$ to $t-31$. The model learns how the stock normally acts in the market, and the abnormal return of a day is the difference between the actual return and what the model predicts. We set up the 30-day gap to keep the estimation window away from pre announcement price movements. To compute the cumulative abnormal returns (CAR), we set up 3 event windows in advance: $[-10, -2]$ to test information leakage, $[0, +1]$ to test announcement reaction, and $[+2, +10]$ to test drift.

Due to the results listed, we were not using conventional parametric tests: the mean and median of clinical disclosures have opposite signs. They have a mean of +3.23% and median of -1.37%. 57% of clinical announcements are negative while the mean is positive. This reflects the pattern of clinical stage biotechnology companies: most results are disappointing while a few can spectacularly succeed. This means that the samples are not normally distributed and thus t-statistics are not reliable.

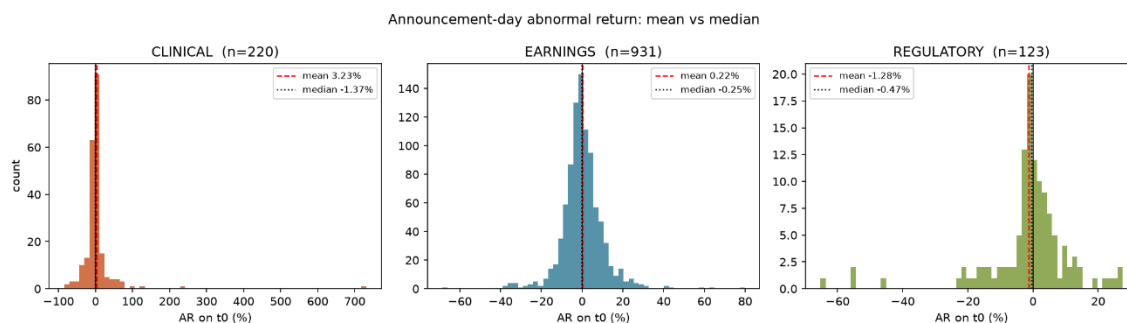


Figure 1. Mean & Median of Announcement-Day Abnormal Return

Second, zero is the wrong null hypothesis. We constructed a placebo benchmark, drew 20 non-event dates for each real event, matched on the same stock and the same calendar month, and running the identical pipeline. If a pattern appears on randomly chosen dates as well, it does not come from the news. The clinical-stage life science companies showed a median CAR of -2.63%, and thus shows that they undergo price decline in regular days.

As a result, we applied empirical p-values by resampling the placebo distribution at the real sample size, together with bootstrap confidence intervals and sign tests. The primary tests were decided as: information leakage, post-announcement drift and clinical-earnings

difference. We also applied a validity check on the announcement day to ensure that we captured real events instead of a hypothesis.

Results

The results show that clinical announcements produce a clear day-zero reaction. The median abnormal return is -1.37% while a placebo baseline is -0.23%. The empirical p-value was below 0.001, indicating the samples are capturing real events.

By analyzing CAR over window, we discovered none of the three primary tests reject:

1. The preannouncement CAR is -0.49% while a placebo baseline is -2.63%. This shows that clinical stage firms decline less before an announcement than regular days.
2. Post-announcement drift is -1.93% while a baseline is -1.80%. This is indistinguishable and thus don't prove post announcement drift.
3. The clinical-earnings difference in drift is -1.18% with a permutation p-value of 0.194. Prices move on the announcement and then stop.

Window	Clinical Actual	Clinical Placebo	p	Earnings Actual	Earnings Placebo
CAR [-10, -2]	-0.494	-2.625	0.057	-1.009	-0.859
AR (0)	-1.367	-0.226	0.000	-0.248	-0.201
CAR [+2, +10]	-1.931	-1.796	0.870	-0.755	-0.097

Table 1. Median Cumulative Abnormal Returns by Window

The null remained valid after five robustness checks. We dropped the single largest observation (a +731% increase in single day), dropped events with negative estimated beta, split the events at 2022, dropped the five most extreme baselines, and focused on the 1% - 99% percentile. Out of all of these tests, none of the primary tests became more significant, and the p-values varied from 0.26 to 0.90.

Limitations

There are several limitations of this event study, listed below:

1. **Survivorship:** The company universe is built based on the companies that are still listed in August 2026, and price data are from yfinance. However, there were multiple life sciences companies that were acquired during the sample period, which is usually because they showed strong market performances. This makes our company universe biased because the surviving samples are usually the ones without the strongest outcomes
2. **Single Cluster:** The company universe is based on the companies from San Diego County. As a result, the data describes the patterns of companies from one specific geography and industry concentration. Since the event study doesn't

- compare to the life sciences companies from San Francisco nor Boston, a general conclusion cannot be reached.
3. **Classification Error:** Since classification hugely relied on LLM, errors are unavoidable. The precision of categorization was only 0.82, and this indicates roughly one out of five events will be classified differently by manual review, and this may contribute to the null result.
 4. **The Joint Hypothesis Problem:** Any test of market efficiency is simultaneously a test of the model that's used to compute the expected returns. While the null result may mean prices are efficient, it also may mean that single-factor market model is wrongly specified for the life sciences companies. Our placebo test partly addressed this problem since it is applied for non-event dates.

Conclusions

This paper questioned whether the market absorbs clinical trials results more slowly compared to regular financial reports, due to that trial results are highly professional and require more knowledge to be understood. By analyzing 2,349 events from 60 San Diego life sciences companies, we found no evidence that it does. On the same day the clinical trials are released, the price is changed, and it moves only once. We didn't detect leakage of information 10 days ahead, nor drift 10 days after the events.

To prove the effectiveness of the null, we applied placebo benchmark, which showed that clinical-stage life sciences companies go through a 2.6% decline over a nine days window, so testing with a zero baseline would have produced a leakage finding. Besides that, the results went through five robustness checks, and the p-values vary between 0.26 to 0.90.

The extensions of this study, as stated previously, include geographical control groups and price data that covers delisted companies. With these extensions, a more general and representative conclusion can be reached.